SBA-Route: Domain-Based Routing for Scalable and Low-Latency 5G Distributed Control Planes

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***Abstract*—The transition to distributed 5G/6G architectures reveals a fundamental gap: the Service-Based Architecture (SBA) lacks a scalable routing substrate, directly degrading end-to-end Quality of Service (QoS) for latency-sensitive mobile applications. While the Network Repository Function (NRF) handles service discovery, 3GPP standards provide no routing protocol for information propagation across distributed NRFs. This omission causes inefficient edge-to-edge discovery that inflates handover latency, quadratic configuration complexity, and network-wide routing instability. Since no routing protocol exists for the distributed SBA control plane, existing approaches resort to non-routing workarounds such as caching or Network Function (NF) replication, which cannot fundamentally resolve these issues. We propose SBA-Route, a domain-based routing framework for the SBA discovery substrate. By organizing NRFs into domains with intra-domain flooding and inter-domain aggregated advertisement, SBA-Route enables direct edge-to-edge discovery and reduces peering complexity from $O(N^2)$ to $O(N)$. Its route aggregation and damping mechanisms further suppress instability. Evaluations demonstrate that SBA-Route reduces inter-domain handover latency by up to 76.6%, lowers central NRF CPU usage by 74%, and suppresses routing oscillations by 71.7%, significantly improving QoS for mobile users in distributed deployments.**

***Index Terms*—5G/6G networks, Quality of Service, distributed control plane, edge computing, NRF, routing protocol**

I. Introduction

The 5G SBA represents a fundamental shift in mobile network design, enabling flexible, distributed deployment of NFs across geographically dispersed locations [1], [2]. This architectural flexibility has proven particularly valuable for User Plane Functions (UPFs), which operators now routinely deploy at edge sites to minimize data plane latency [3]–[5] for latency-sensitive applications such as autonomous vehicles [6]–[8], virtual and augmented reality [9], and low-latency edge inference [10]–[12]. For these applications, even tens of milliseconds of additional handover latency can cause visible frame drops in Extended Reality (XR) streaming, unsafe control gaps in autonomous driving, or stale inference results at the edge, making control plane discovery latency a direct QoS bottleneck. Looking forward, the industry envisions fully-distributed 6G architectures where control plane functions themselves are deployed across thousands of sites to support emerging ultra-low-latency and high-mobility scenarios.

However, as networks evolve toward distributed control plane architectures, a fundamental architectural gap becomes apparent: *the SBA service discovery substrate lacks a scalable routing framework for information propagation*. While 3GPP standards [1], [2] define the NRF [13] for service registration and discovery, they provide no routing protocol governing how distributed NRFs should exchange and propagate service reachability information. The standards implicitly assume either full-mesh connectivity (which incurs $O(N^2)$ scaling complexity) or slice-level partitioning, both only viable when NRFs remain centralized. This gap is analogous to defining IP packet forwarding without providing a routing protocol like the Border Gateway Protocol (BGP) [14]. Without such a protocol, the system fundamentally cannot scale as a cohesive whole.

This architectural void manifests acutely in today's edge control plane deployments. The research community has explored deploying control plane NFs at edge locations to reduce control plane latency [4], [15], [16], and the industry has proposed distributed control plane architecture as a fundamental design principle for 6G [17]–[21]. However, both research prototypes and industry proposals immediately encounter three systemic challenges that directly degrade QoS: edge-to-edge NF discovery requiring traversal through distant central NRFs (adding 24-178ms latency [22]), inflating handover delays and disrupting service continuity; $O(N^2)$ configuration complexity with full-mesh NRF topologies; and network-wide routing instability as NF state changes propagate globally. Crucially, since no routing protocol exists for the distributed SBA control plane, existing works can only resort to non-routing workarounds, such as caching [15] or NF replication [4], which cannot fundamentally resolve these interconnected challenges.

To address these challenges, we present SBA-Route, a domain-based routing framework that provides the missing routing substrate for distributed SBA deployments. SBA-Route

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organizes NRFs into logical domains, employing intra-domain flooding for local reachability and inter-domain aggregation for global scalability. This design enables direct edge-to-edge NF discovery without centralized intermediaries, effectively transforming the inefficient V-shaped discovery path into a direct one. In contrast to existing 3GPP-compliant multi-NRF deployments, which rely on static interconnection choices such as full-mesh peering or hierarchical registration, SBA-Route introduces an explicit routing mechanism with loop prevention, aggregation, and damping to contain churn and suppress instability. We implement SBA-Route on the open-source free5GC platform and validate its QoS improvements in realistic edge scenarios.

In summary, we make the following contributions:

- **Scalable Domain-Based Routing Protocol:** We design an SBA-native routing protocol that replaces $O(N^2)$ full-mesh connectivity with $O(N)$ neighbor peering. By combining intra-domain flooding and inter-domain aggregation with sequence-number-based loop prevention, we ensure efficient information propagation across distributed NRFs.
- **Route Aggregation and Damping:** We implement NRF Speaker-based route aggregation and damping mechanisms that confine NF state changes within local domains, absorbing transient churn at domain boundaries and improving network-wide stability.
- **Comprehensive Evaluation:** We implement SBA-Route on free5GC and evaluate it against the 3GPP standard and existing non-routing approaches. Results show that SBA-Route reduces inter-domain handover latency by up to 76.6%, lowers central NRF CPU usage by 74%, reduces routing table growth by 83%, and suppresses routing oscillations by 71.7%.

II. Background

A. 5G Core Network Architecture

The 5G core network adopts an SBA that separates control plane and user plane functions. As illustrated in Figure 1, the user plane handles data traffic through the UPF, increasingly deployed at edge locations to minimize latency. The control plane traditionally remains centralized in cloud data centers.

The *Access and Mobility Management Function (AMF)* manages User Equipment (UE) registration, mobility, and connection via the N2 interface. The *Session Management Function (SMF)* handles session establishment and configures user plane paths through the N4 interface. The *NRF* serves as the service registry for all NFs: each NF registers its NF Profile (service capabilities, capacity, location, status), and the NRF enables other NFs to discover appropriate service providers through query-based mechanisms.

The NRF also monitors NF health through periodic heartbeat exchanges. When an NF fails to respond, the NRF marks it as unavailable and propagates this status update. When multiple NRFs exist across different regions, they interconnect to exchange routing information for cross-region NF discovery.

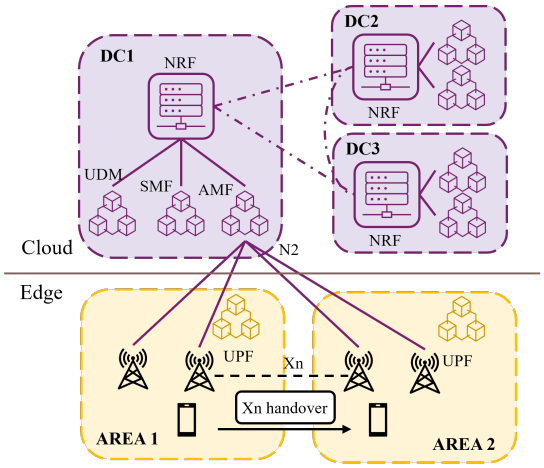


Fig. 1: Traditional 5G deployment: control plane NFs reside in centralized cloud data centers, while UPFs are deployed at edge sites.

B. Handover Procedures

User mobility triggers different handover procedures depending on the network topology. When a UE moves between next-generation Node Bs (gNBs) served by the same AMF, the network performs an Xn handover with minimal discovery overhead. However, when a UE crosses AMF service boundaries, the network must perform an N2 handover: the source AMF must consult the NRF to discover an appropriate target AMF based on the UE's new location. In edge deployments (Figure 2), this NRF-mediated discovery introduces significant additional latency, as the NRF resides in a distant cloud data center. This discovery overhead directly degrades handover QoS, increasing service interruption time for latency-sensitive applications [4], [23].

C. Routing Propagation

The 3GPP standard defines the NF Profile structure for each NF. Each NRF appends the NF Info of all registered NFs into its own NRF Profile, advertising it as routing information. NRFs may interconnect and exchange NRF Profiles for cross-domain discovery, but the specification deliberately leaves the detailed routing exchange behavior as implementation-dependent.

Since the standard specifies no routing protocol for multi-NRF interconnection and provides no loop prevention mechanism, operators typically choose full-mesh topologies and avoid forwarding neighbors' NRF Profiles to prevent loops.

An alternative is *hierarchical NRF deployment*, where edge NRFs register with central NRFs in a star topology. Each edge NRF sends its NRF Profile, containing NF Info for all locally registered NFs, to the central NRF, which merges this information into its own NRF Profile. Central NRFs maintain full-mesh connectivity among themselves and exchange their enlarged NRF Profiles. This reduces edge-to-central connectivity to $O(N)$, but cross-region discovery still requires traversing the central tier, preserving the V-shaped latency problem.

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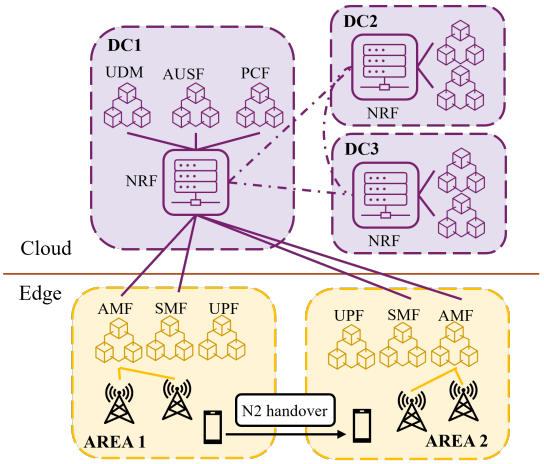


Fig. 2: Edge control plane deployment with centralized NRF discovery: during N2 handover between edge areas, the discovery path must traverse from edge to cloud and back, creating the V-shaped latency problem.

The absence of a routing protocol for multi-NRF interconnection is the root cause of the challenges we identify next.

III. Problem Analysis and Challenges

Distributed deployment of the control plane introduces three fundamental challenges that expose critical architectural gaps in the current SBA design. We use real-world mobility handover procedures as a running example, a critical QoS-sensitive use case where control plane functions must coordinate across geographical locations for seamless user mobility.

A. Challenge 1: Edge-to-Edge Discovery Path Latency

Prior works [4], [15] have proposed deploying AMFs and SMFs at the edge to reduce control plane latency. However, the reduced service coverage area of each edge AMF dramatically increases the frequency of N2 handovers, each requiring NRF-mediated target AMF discovery.

As illustrated in Figure 2, when edge-deployed AMFs must discover other edge-deployed AMFs, centralized NRF architectures create a V-shaped discovery path. The source AMF must query a distant cloud NRF despite the target AMF being geographically close. This path adds 24-178ms round-trip latency [22], directly degrading handover QoS and substantially offsetting the latency benefits of edge deployment. This challenge generalizes beyond handovers: emerging 6G scenarios such as UE-to-NF discovery [24], [25] and dynamic service composition [26], [27] share the same pattern: *edge entities need to discover other edge entities without traversing distant intermediaries*.

B. Challenge 2: NRF Configuration Complexity

As the number of control plane functions grows, deploying additional NRFs becomes essential due to increasing heartbeat management pressure. As shown in Figure 3 (measured with a 10s heartbeat interval [29]), CPU overhead grows proportionally with heartbeat connections, reducing the number of UEs

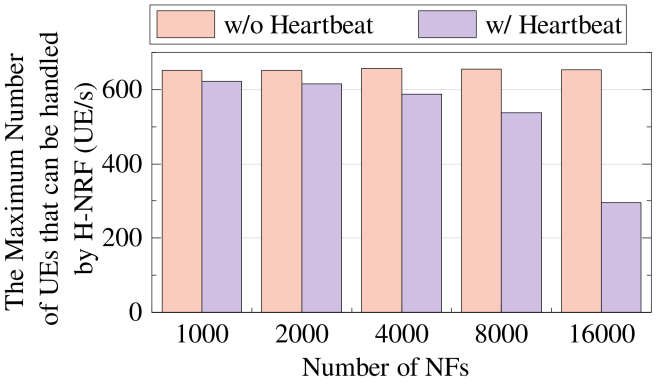


Fig. 3: Maximum UEs served by H-NRF [28] as heartbeat connections increase.

that can be served. A natural response is to deploy additional NRFs across edge sites, but current standards mandate full-mesh connectivity for routing information exchange, requiring $O(N^2)$ connections, which is unacceptable at edge scale.

C. Challenge 3: Network-Wide Routing Oscillations

Under existing approaches, any NF status change triggers routing updates that propagate throughout the entire network. Edge nodes exhibit significantly higher failure rates than centralized core nodes [30]. For N edge nodes with mean failure rate λ, the aggregate routing oscillation frequency is $f_{\text{oscillation}} = N \cdot \lambda$, scaling linearly with edge node count. Each oscillation consumes bandwidth, requires processing at every NRF, and may trigger unnecessary service reselection, further degrading QoS.

D. Why Existing Solutions Fall Short

Three intuitive approaches fall short when examined systematically. **Edge NRF Deployment** partially addresses Challenge 1 but severely exacerbates Challenge 2 ($O(N^2)$ full-mesh) and fails to address Challenge 3. **Hierarchical NRF Deployment** organizes NRFs into a two-tier star topology where edge NRFs register with central NRFs. This reduces edge connectivity to $O(N)$ (Challenge 2), and edge NRFs can resolve intra-region queries locally. However, inter-region discovery still requires traversing the central NRF, preserving the V-shaped latency problem (Challenge 1). Moreover, central NRFs store raw, unaggregated NF profiles from all edge NRFs, and lack damping mechanisms, leaving Challenge 3 unaddressed. **Slice-Based NRF Partitioning** conflicts with geographical distribution requirements, as slices group NFs by services rather than locations.

Notably, all these approaches are architectural workarounds rather than routing protocols. While recent work such as H-NRF [28] improves NRF throughput under large NF counts, it cannot eliminate the inherent heartbeat overhead that scales with the number of managed NFs. Since no routing protocol exists for the distributed SBA control plane, the community has been limited to non-routing solutions, including caching [15], replication [4], and hierarchical deployment,

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none of which can simultaneously enable direct edge-to-edge discovery, reduce complexity to $O(N)$, and suppress routing oscillations. This motivates SBA-Route's integrated domain-based routing framework.

IV. SBA-Route Design

SBA-Route introduces a domain-based routing framework that provides the missing routing substrate for distributed SBA deployments. We make the following assumptions: (1) edge sites have IP connectivity to neighboring edge sites and to the central core; (2) domain boundaries are configured by the operator based on geographical or administrative criteria; (3) NRFs within a domain can reach at least one other NRF in the same domain (no full-mesh required). The framework addresses distributed control plane challenges through three core components: distributed NRF deployment, domain routing protocol, and NRF Speaker route aggregation with damping. Figure 4 illustrates the architecture, comprising a Central Domain and multiple Edge Domains. Each domain is equipped with Speaker NRFs for external communication. Every Speaker NRF implements route aggregation and damping on inter-domain connections. Within each domain, NRFs facilitate local service discovery via controlled flooding with sequence-number-based loop prevention. Inter-domain routing includes default routes from the Central Domain and direct edge-to-edge announcements between Edge Domains. While we present the design in the context of edge computing, SBA-Route is applicable to any distributed control plane architecture.

A. Design Rationale

A natural question arises: why not use existing distributed databases (e.g., etcd [31], Consul [32]) or IP routing protocols (e.g., BGP)?

Distributed key-value stores like etcd prioritize consistency over availability (CP systems) and incur significant synchronization overhead that scales poorly with edge churn rates. NRF data exhibits high locality and requires high availability (AP system) rather than strict global consistency. Meanwhile, Multiprotocol BGP (MP-BGP) [33] remains fundamentally a reachability protocol rather than a service discovery mechanism. Adapting BGP would require a complex dual-mapping layer to translate the 3GPP NRF Profile's rich JSON-based attributes into BGP Network Layer Reachability Information (NLRI) and back. SBA-Route instead leverages the native NRF Profile structure directly, borrowing BGP's proven stability mechanisms (aggregation and damping) but implementing them natively within the SBA's HTTP/2 application layer.

B. Domain-Based NRF Architecture

SBA-Route co-locates NRF instances with AMF/SMF clusters at edge sites and co-designs distributed deployment with a domain-based routing protocol to avoid quadratic complexity.

As illustrated in Figure 4, SBA-Route organizes NRFs into Central and Edge Domains. In Edge Domains, NRFs are loosely coupled via intra-domain neighbor connections with controlled flooding. Speaker NRFs serve as gateways. The Central Domain's Speaker NRFs provide default routes and aggregate global reachability. The synergy between **intra-domain NRF interconnection** and **direct Edge-to-Edge Routing Announcements** jointly eliminates the central bottleneck, enabling discovery without traversing the central core. This transforms the V-shaped discovery path into a direct one, reducing handover latency by an order of magnitude.

C. Domain Routing Protocol

SBA-Route's domain routing protocol organizes NRFs into logical partitions, transforming configuration complexity from $O(N^2)$ to $O(N)$ and confining routing updates within domain boundaries.

1) Domain Definition: SBA-Route organizes the network into routing domains, where each domain contains multiple NRF instances. Domains typically correspond to geographical regions or administrative boundaries. The specific criteria for domain formation and the number of NRFs per domain are deployment decisions outside our scope. Importantly, domains serve to control failure propagation and routing complexity rather than restrict connectivity; NFs in different domains remain mutually reachable through the inter-domain routing mechanism.

2) Intra-Domain Routing: Unlike IP routing which requires path information, NRF routing only needs to ensure that NF Profiles reach all relevant NRFs, since profiles contain direct connectivity information (IP addresses or Fully Qualified Domain Names (FQDNs)). SBA-Route implements intra-domain routing through controlled flooding with loop prevention. This mechanism propagates both locally registered NF profiles (intra-domain routes) and external reachability information injected by Speaker NRFs (inter-domain routes) throughout the domain.

When a new NRF joins a domain, it connects to one or more existing NRFs, there is no requirement for full-mesh connectivity or specific topology. The new NRF registers with its chosen neighbors using the standard NF registration procedure, with one extension: each registration includes a sequence number (seqNum) that strictly increments with each update from that NRF. The new NRF maintains heartbeat exchanges only with these direct neighbors.

Direct neighbor NRFs propagate the new NRF's profile to their other neighbors, including the seqNum. When an NRF receives such an indirect registration, it checks whether it already has a profile for that NfInstanceID with a seqNum greater than or equal to the received seqNum. If so, the routing update terminates at that NRF without further propagation. Otherwise, the NRF stores the profile and continues propagating it to its neighbors.

This mechanism ensures that each NF Profile eventually reaches all NRFs within the domain while preventing routing loops and unnecessary duplicate propagations. Routing updates and deletions follow the same protocol. If a directly connected NRF fails to respond to heartbeat requests, its

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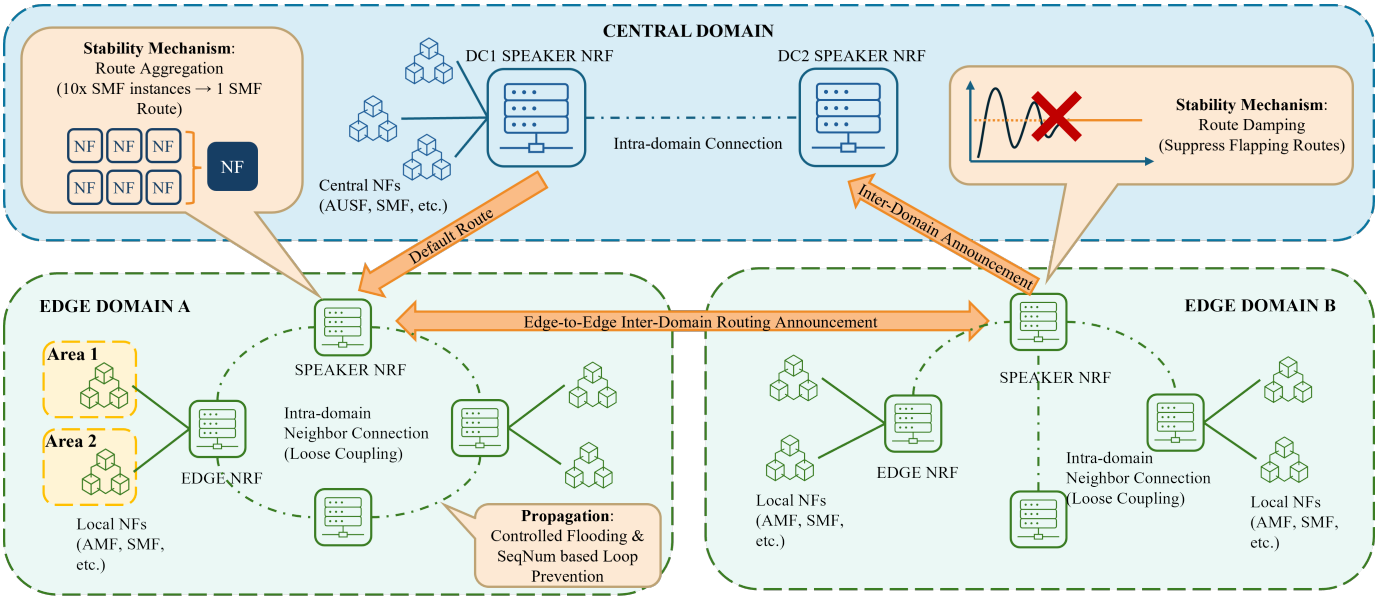


Fig. 4: SBA-Route architecture: Speaker NRFs implement Route Aggregation and Route Damping at domain boundaries. Within each domain, NRFs propagate routes via controlled flooding with loop prevention. Intra-domain interconnection and direct Edge-to-Edge Routing Announcements jointly eliminate central bottlenecks.

neighbor sends a deregistration message with that NRF's most recent seqNum, triggering removal throughout the domain.

3) Inter-Domain Routing: Inter-domain routing follows three key rules: (1) domain NRF Speakers propagate all NF Infos from their domain to other domains; (2) core network domains advertise only a default route toward edge domains, both simplifying edge routing tables and preventing compromised edge domains from mapping the core topology; (3) domains interconnect through designated NRF Speakers.

To address scalability at central hubs, SBA-Route allows central domains to deploy multiple active Speaker NRFs. This is viable because southbound speakers only advertise default routes (no conflicts), and northbound inter-domain routes from different edge domains merge without conflict. Speaker NRFs in Edge Domains can also establish direct peering for Edge-to-Edge Routing Announcements, bypassing the central domain. Routing propagation between Speakers follows the same sequence-number-based mechanisms as intra-domain routing.

D. Stability Mechanisms: Route Aggregation and Damping

Every Speaker NRF implements route aggregation and damping at domain boundaries.

Route Aggregation. Speaker NRFs aggregate fine-grained local NF instances into concise service routes before announcing them to other domains. When individual NF instances fail or recover, the aggregated route remains stable as long as at least one equivalent instance remains available, absorbing the vast majority of intra-domain routing changes at the domain boundary.

Route Damping. SBA-Route employs penalty-based route damping inspired by BGP route flap damping. If a route oscillates too frequently, the Speaker suppresses its advertisement until it stabilizes, preventing routing instability from cascading across domains.

V. Implementation

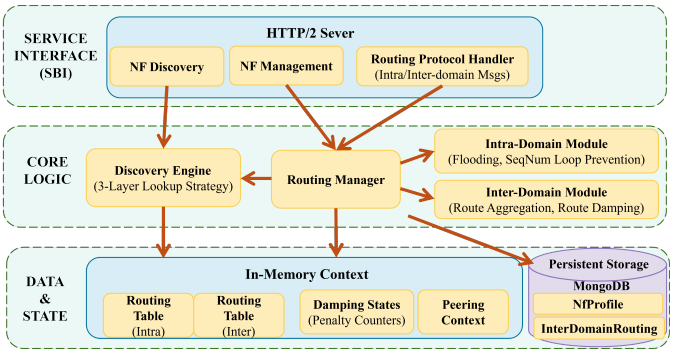


Fig. 5: SBA-Route NRF Internal Architecture: Service Interface, Core Logic, and Data & State layers.

We implement SBA-Route on top of free5GC (v4.1.0) [34], an open-source 5G core platform written in Go. The modifications span approximately 6,000 lines of code across 24 files, primarily within the NRF module, while maintaining full backward compatibility with standard 3GPP interfaces.

As illustrated in Figure 5, we redesigned the NRF into three layers. The **Service Interface** extends the HTTP/2 server with a Routing Protocol Handler for intra- and inter-domain routing messages. The **Core Logic** centers on a Routing Manager orchestrating the Discovery Engine (with local, intra-domain, and inter-domain lookup) and two specialized modules for intra-domain flooding and inter-domain aggregation/damping. The **Data & State** layer maintains in-memory routing tables,

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damping states, and peering context, asynchronously synchronized with MongoDB for durability.

We extended the `Nnrf_NFManagement` API with route flooding endpoints using strictly increasing sequence numbers for loop prevention, and custom HTTP headers (e.g., `X-Cross-Domain-Query`) for cross-domain discovery cycle prevention. Route aggregation groups NF instances by `NF Type` into single `NrfInfo` entries before inter-domain propagation. Route damping uses a penalty-based system inspired by BGP Route Flap Damping (RFC 2439), tracking stability at `(DomainId, NfType)` granularity with exponential decay.

VI. Evaluation

A. Experimental Setup

We evaluate SBA-Route using emulation-based experiments on free5GC with scaled simulations for larger topologies. Our testbed consists of 20 edge sites distributed across a metropolitan area, each equipped with an SBA-Route NRF, 25 AMF and SMF instances, and UPF nodes, with each edge site generating approximately 50 discovery requests per second. In the SBA-Route deployment, although the protocol supports arbitrary connectivity, we constructed a balanced diameter-2 graph with balanced node degrees to avoid hotspots. We applied this topology for both intra-domain NRF connections and inter-domain peering. We emulate UE mobility using UERANSIM. We configure the intra-domain heartbeat interval to 3s and the inter-domain interval to 10s to balance freshness and overhead. The damping parameters are configured to trigger suppression after 3 consecutive oscillations. For centralized baselines, we deploy central domains on AWS (avg round-trip time (RTT) 34.14 ms).

1) Baselines: We compare SBA-Route against the 3GPP standard, a hierarchical NRF deployment, and existing non-routing workarounds. We additionally use ablation variants to isolate the contribution of each mechanism:

- **3GPP Standard (Centralized)**: Traditional architecture with full-mesh central NRFs, forcing V-shaped discovery paths.
- **Hierarchical NRF**: Two-tier star topology where edge NRFs register their NRF Profiles with central NRFs. Central NRFs merge received NF Info into their own NRF Profiles and maintain full-mesh connectivity among themselves. Edge NRFs resolve local queries directly; cross-region queries are forwarded to the central tier.
- **3GPP w/ Stability**: Centralized architecture enhanced with our stability mechanisms to isolate their benefits independently of distributed routing.
- **SBA-Route w/o Stability**: Our distributed framework without aggregation/damping to quantify the contribution of stability mechanisms.
- **D&C** [15]: A caching-based approach (non-routing) that optimizes central discovery by caching results at the edge.
- **CellClone** [4]: A replication-based approach (non-routing) deploying multiple edge NF replicas to minimize latency.

2) Metrics: We focus on four QoS-relevant aspects:

- **End-to-End Handover Latency**: Total duration of N2 handovers, directly reflecting user-perceived QoS.
- **Service Discovery Latency**: Time for NRFs to resolve queries, a key contributor to control plane QoS.
- **Scalability**: CPU/memory usage and routing table growth as the network scales.
- **Stability & Overhead**: Convergence time, signaling bandwidth, and cross-domain traffic under failure scenarios.

B. End-to-End Handover Latency

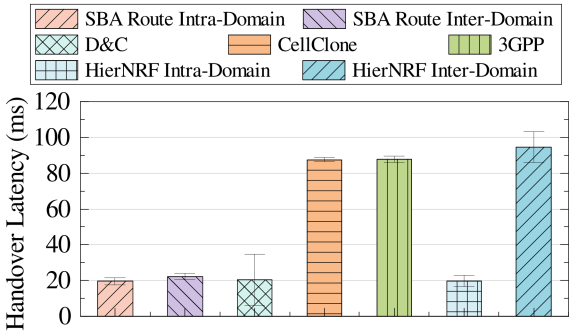


Fig. 6: End-to-end N2 handover latency comparison across different approaches.

We evaluate N2 handover latency to assess the real-world impact of discovery performance (Figure 6).

For *intra-domain* handovers, SBA-Route achieves an average latency of 19.63 ms. This is comparable to Hierarchical NRF (19.81 ms), whose edge NRFs also resolve local queries directly, and D&C (20.41 ms), which serves requests from local cache. However, D&C exhibits significantly higher variance (standard deviation 14.40 ms) because cache misses force it to query the distant central NRF, incurring a heavy latency penalty. SBA-Route avoids this instability by resolving all local queries within the edge domain.

For *inter-domain* handovers, SBA-Route demonstrates its core advantage. By enabling direct edge-to-edge discovery, it achieves 22.11 ms average latency. In contrast, the 3GPP Standard (87.85 ms), CellClone (87.56 ms), and Hierarchical NRF (94.52 ms) all suffer from the V-shaped routing path, where discovery requests must travel to the central cloud and back. Although Hierarchical NRF deploys edge NRFs, cross-region queries still require central NRF mediation, yielding latency even higher than the centralized baselines. SBA-Route eliminates this detour, reducing inter-domain latency by 76.6% over Hierarchical NRF, 74.8% over the 3GPP Standard, and 74.7% over CellClone. This 4× latency reduction is critical for supporting high-mobility scenarios in future 6G networks, where frequent handovers would otherwise degrade user experience.

C. Network Scalability

We evaluate the scalability of SBA-Route by comparing its resource consumption and discovery performance against

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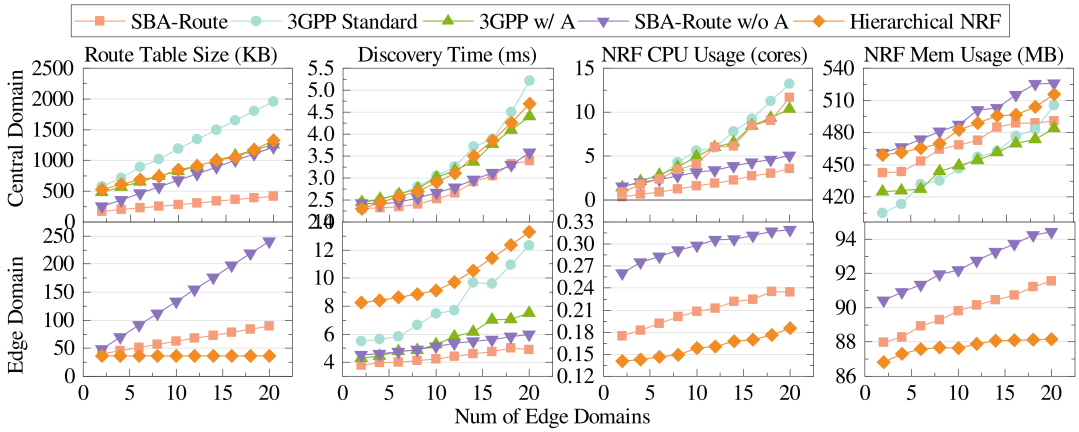


Fig. 7: Scalability comparison of Route Table Size, Discovery Time, CPU Usage, and Memory Usage across Central and Edge domains. The horizontal axis represents the number of Edge Domains. The 3GPP and 3GPP w/ A baselines lack Edge NRF data as they rely on a centralized architecture.

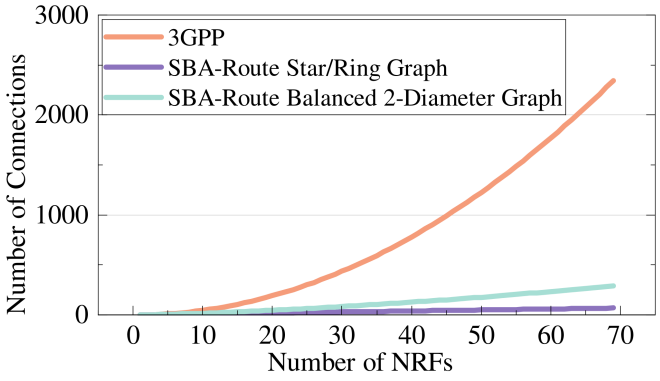


Fig. 8: Comparison of NRF interconnection complexity. 3GPP requires full-mesh connectivity ($O(N^2)$), while SBA-Route supports flexible topologies like Star/Ring ($O(N)$) or Balanced Diameter-2 Graph ($O(N^{1.5})$).

the 3GPP baseline and Hierarchical NRF as the network scales. Figure 7 presents the results for both Central and Edge domains. Since D&C shares the same architecture as the 3GPP standard, its scalability performance is essentially identical. Conversely, CellClone requires multiple NF replicas, resulting in significantly larger routing tables and higher heartbeat overhead compared to the 3GPP baseline, leading to much lower scalability. To maintain chart clarity, these results are not included in the figure.

Central Domain Performance. In the central domain, SBA-Route demonstrates significant efficiency gains. The route table size growth rate of SBA-Route is reduced by 83% compared to the 3GPP baseline. Hierarchical NRF's central NRFs store raw, unaggregated NF profiles merged from all edge NRFs, resulting in route table sizes slightly lower than the 3GPP baseline but following the same linear growth trend. For discovery time, SBA-Route maintains stable sub-5ms latency, while both 3GPP and Hierarchical NRF exhibit comparable but higher latency that grows with the number of edge domains. In terms of computational overhead, SBA-Route reduces central NRF CPU usage by 74% compared to the 3GPP baseline. Hierarchical NRF's central CPU usage is comparable to the 3GPP baseline, as the overhead of profile merging from edge NRFs offsets the benefit of reduced full-mesh connectivity. Memory consumption follows a similar pattern across 3GPP and Hierarchical NRF, while SBA-Route maintains a lower and more stable footprint.

Edge Domain Performance. At the edge, all approaches with edge NRFs (SBA-Route, SBA-Route w/o A, and Hierarchical NRF) maintain minimal resource footprints for route table size, CPU usage, and memory usage, meeting the stringent requirements of edge computing environments. The key differentiator is edge discovery time: SBA-Route achieves stable low latency by resolving queries locally or through direct edge-to-edge peering. In contrast, Hierarchical NRF's edge NRFs can only resolve queries for locally registered NFs; cross-region queries must be forwarded to the central tier, causing edge discovery latency to grow significantly as the number of domains increases.

Impact of Aggregation. The results in Figure 7 further highlight the critical role of aggregation. Route aggregation effectively compresses route table entries for both 3GPP (when enhanced with our mechanisms) and SBA-Route. This compression directly translates to reduced CPU usage during discovery operations, which in turn lowers the overall discovery time.

Connection Complexity. Figure 8 illustrates the scalability of NRF interconnection. Traditional 3GPP standards mandate a full-mesh topology, resulting in $O(N^2)$ connection complexity. Hierarchical NRF reduces this to $O(N)$ via a star topology where each edge NRF connects to a central NRF, but all traffic must traverse the central tier, creating a single point of congestion. SBA-Route also achieves $O(N)$ with a star or ring topology, but distributes traffic across peers without a central bottleneck. In our evaluation, we utilized a balanced

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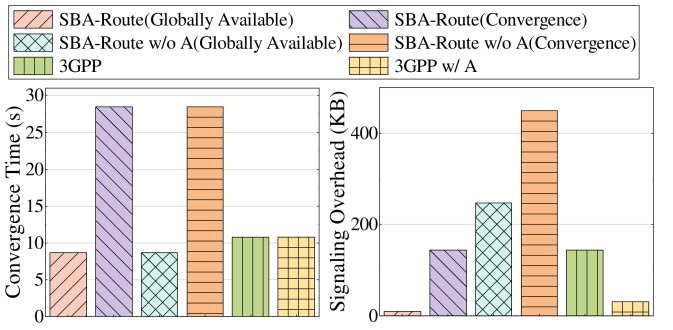
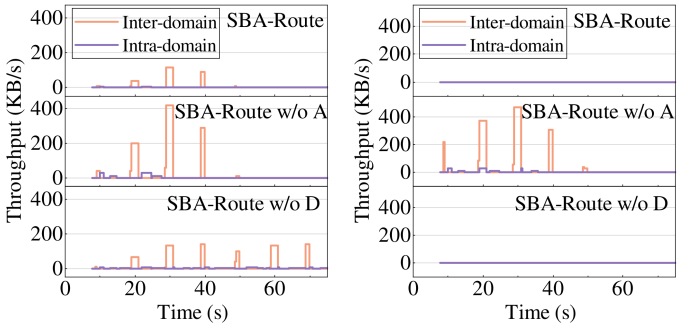


Fig. 9: Routing convergence performance. (a) Convergence time. (b) Signaling overhead.



(a) AMF Flapping (b) SMF Flapping

Fig. 10: Cross-domain routing traffic under AMF and SMF flapping scenarios.

diameter-2 graph ($O(N^{1.5})$) to avoid hotspots while ensuring bounded path lengths.

D. Routing Convergence and Overhead

Figure 9(a) compares convergence times. SBA-Route achieves *Global Availability* (reachability via the Central Domain) with latency comparable to the 3GPP baseline. While reaching the fully *Converged* state (enabling direct edge-to-edge discovery) takes longer due to multi-hop propagation across domains, this does not compromise service availability. During this transient period, discovery requests simply fall back to the default route via the Central Domain, ensuring a "fail-soft" behavior.

Figure 9(b) analyzes signaling overhead. Unlike the 3GPP baseline which concentrates traffic on central links, SBA-Route distributes signaling across dispersed inter-domain links, avoiding bottlenecks. Furthermore, our aggregation mechanism reduces total signaling overhead by 65%–96% compared to non-aggregated approaches, validating its critical role in enabling scalable distributed routing.

E. Routing Stability under NF Churn

We evaluate stability by toggling an NF online/offline every 7s to simulate failure. Figure 10 demonstrates how our two stability mechanisms handle different scenarios:

Scenario 1: Unique AMF (Damping). In Figure 10a, the flapping NF is a unique instance (e.g., specific Tracking Area Identity (TAI) coverage). Since its state change must be propagated, aggregation cannot hide it. Here, SBA-Route's *damping* mechanism kicks in, suppressing the oscillating route after the 4th cycle. This limits peak cross-domain traffic to 118 kb/s. Hierarchical NRF, lacking any damping mechanism, behaves similarly to the undamped baseline: each edge NRF state change triggers an NRF Profile update to the central tier, and the central NRF propagates the enlarged profile to all other central NRFs, causing oscillations to persist indefinitely. In contrast, without aggregation, the peak traffic surges to 417 kb/s, and without damping, oscillations persist indefinitely, flooding the network. Note that this paper focuses on the functionality of the damping mechanism; specific parameter selection depends on operator requirements and is out of scope.

Scenario 2: Redundant SMF (Aggregation). In Figure 10b, the flapping NF is one of multiple redundant instances. Here, SBA-Route's *aggregation* completely masks the local churn. Because the domain's overall service capability remains unchanged despite one instance failing, the Speaker NRF does not generate any inter-domain updates. This results in zero cross-domain traffic, perfectly isolating the failure within the local domain. Hierarchical NRF, however, performs raw merging without aggregation: each individual NF instance change causes the edge NRF to update its NRF Profile and re-register with the central NRF, generating cross-domain traffic even when the overall service capability is unchanged.

VII. Discussion

A. Domain Formation Sensitivity

SBA-Route's performance depends on how NRFs are grouped into domains. A natural concern is sensitivity to sub-optimal domain choices. We analyze two dimensions: domain granularity and domain boundary placement.

Granularity. Finer-grained domains (fewer NRFs per domain) reduce intra-domain flooding scope and improve failure isolation, but increase the number of inter-domain peering sessions and the volume of aggregated advertisements. Coarser domains amortize inter-domain overhead but widen the blast radius of local churn. In our evaluation, domains of 3–5 NRFs strike a practical balance: intra-domain flooding converges within milliseconds, while aggregation absorbs the majority of local state changes at domain boundaries. Importantly, SBA-Route degrades gracefully at both extremes: a single-NRF domain reduces to pure inter-domain routing, while a single global domain reduces to flat flooding with no aggregation, equivalent to a baseline distributed NRF deployment.

Boundary placement. Domains aligned with geographical or administrative boundaries (e.g., metropolitan areas, operator regions) naturally match mobility patterns, since most handovers occur within a region. When domain boundaries bisect high-mobility corridors, inter-domain handovers increase, but direct edge-to-edge peering between adjacent domains mitigates the latency impact. Operators can further tune boundaries

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using mobility statistics from existing network analytics (e.g., Network Data Analytics Function (NWDAF) [35]) without modifying SBA-Route itself.

B. Backward Compatibility and Incremental Deployment

SBA-Route is designed for incremental adoption. All routing extensions use standard HTTP/2 endpoints on the existing `Nnrf_NFManagement` service, requiring no changes to non-NRF network functions. An operator can deploy SBA-Route NRFs alongside legacy NRFs: legacy NRFs simply do not participate in flooding or aggregation, and SBA-Route NRFs treat them as leaf nodes with static registrations. This allows operators to upgrade individual sites without a network-wide flag day, which is critical for production 5G deployments where service continuity is paramount.

C. Broader QoS Implications

While our evaluation focuses on N2 handover latency as the primary QoS metric, SBA-Route's benefits extend to other QoS-sensitive control plane operations. Session establishment (Protocol Data Unit (PDU) session setup) requires SMF discovery via NRF, where the same V-shaped latency problem applies. Network slice selection involves Network Slice Selection Function (NSSF)–NRF interactions that similarly benefit from local discovery. As 6G architectures introduce UE-to-NF direct discovery [24] and dynamic service composition [26], the demand for low-latency, scalable NF discovery will intensify. SBA-Route's domain-based routing substrate provides a general foundation for these emerging QoS requirements, not limited to mobility management alone.

VIII. Related Work

a) Distributed and Edge Control Plane: Several works have explored distributing 5G control plane functions to improve latency and resilience. CellClone [4] replicates NFs at edge sites to reduce handover latency, while D&C [15] caches NF profiles at on-premise sites to mitigate signaling overhead in private 5G. ECHO [36] addresses reliability of distributed cellular cores in public clouds through state replication. These approaches focus on data availability (caching, replication, state management) rather than routing; they assume a centralized or hierarchical NRF can always be reached. SBA-Route addresses the orthogonal problem of *how* distributed NRFs exchange and propagate discovery information, and is complementary to these data-plane optimizations.

b) 3GPP NRF Interconnection and Deployment Practice: Our contribution is also distinct from existing 3GPP NRF interconnection mechanisms and operator deployment practices. The 3GPP SBA specifications define NRF registration, discovery, and the exchange of NRF Profiles across multiple NRFs, but leave the propagation policy, topology construction, and loop handling largely implementation-dependent [1], [2], [13]. In practice, operators and prototypes therefore tend to adopt static designs such as full-mesh peering among central NRFs or hierarchical registration from edge NRFs to a central tier, which provide connectivity but not an explicit routing protocol with distributed path control or churn containment. SBA-Route builds on the standard NRF interfaces, but adds precisely these missing routing-plane functions: neighbor-based propagation, sequence-number-based loop prevention, domain-level aggregation, and damping at domain boundaries.

c) Distributed Naming and Addressing: The need for a more dynamic and distributed mobile core is widely recognized. Recent work on distributed application addressing for 6G [37] shares our goal of breaking hierarchical bottlenecks. It proposes a clean-slate, application-oriented naming and resolution fabric to enable low-latency service access. However, it operates at the application layer, focusing on endpoint naming rather than reforming the routing of 3GPP NF capability information. SBA-Route is complementary, providing a concrete mechanism to efficiently route the control-plane information that 3GPP networks already expose. It employs service-oriented aggregation and stability-aware damping to manage information fidelity, aspects not addressed by the naming-focused approach.

d) Classic Route Summarization: Our aggregation and propagation mechanisms also differ from traditional routing protocols. BGP aggregates IP prefixes based on administrative policy to provide reachability, deliberately discarding service-level attributes for scalability [14]. Link-state protocols like Open Shortest Path First (OSPF) summarize topology metrics within areas [38]. These schemes are ill-suited for multi-attribute service discovery, as their single-key summarization destroys the cross-attribute correlations essential for avoiding phantom routes. In contrast, SBA-Route uses NF profile aggregation to create holistic, multi-attribute summaries that preserve service semantics.

IX. Conclusion

We identify a critical gap in the 5G SBA: the absence of a routing protocol for distributed control planes, which directly degrades QoS for latency-sensitive mobile applications. Since no such routing protocol exists, existing approaches resort to non-routing workarounds that cannot fundamentally resolve the interconnected challenges of discovery latency, configuration complexity, and routing instability. To address this, we present SBA-Route, a domain-based routing framework for the SBA discovery substrate, featuring intra-domain flooding and inter-domain aggregation with route damping. Our evaluation demonstrates that SBA-Route reduces inter-domain handover latency by up to 76.6%, lowers central NRF CPU usage by 74%, and suppresses routing oscillations by 71.7%, significantly improving end-to-end QoS. By providing a scalable, backward-compatible routing substrate, SBA-Route resolves key bottlenecks in current edge deployments and establishes a foundation for future fully-distributed 6G architectures.

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